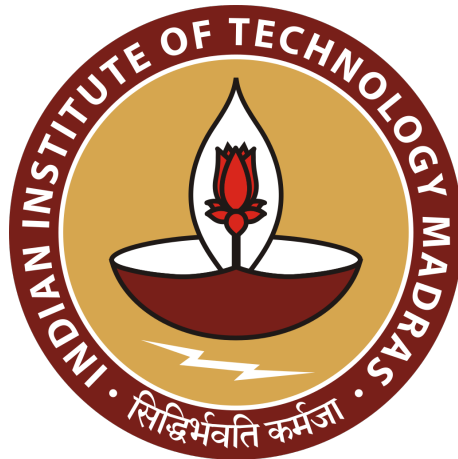


Lab Assignment - 2

Bioinformatics Databases



BT3040 - Bioinformatics

Even Semester 2026

Course Instructor: Professor Michael Gromiha

K.Varunkumar | BE23B016

Department of Biotechnology

Indian Institute of Technology, Madras

Question 1

DNA Data Bank of Japan (DDBJ)

The screenshot shows the DDBJ search interface. On the left, there are filters for Keyword, Database (JGA, BioProject, BioSample, SRA), Type (homo sapiens), Organism (homo sapiens), and Published Date. The main area displays search results for 'Homo sapiens' with 14,907,272 sequences. The results are listed in a table with columns for accession number, sample name, and publication date. The first few results are DRS179359, DRS179360, DRS179361, and DRS179362, all published on 04/12/2021 at 2:15 AM.

Figure 1.1: DDBJ has 14,907,272 sequences under *Homo sapiens*

Genbank

The screenshot shows the GenBank search interface. The top header includes the NIH logo and the text 'National Library of Medicine National Center for Biotechnology Information'. The search bar shows 'Nucleotide' and 'txid9606'. The results are displayed in a table with columns for accession number, sample name, and publication date. The first few results are Homo sapiens BRCA2 DNA repair associated (BRCA2), transcript variant 7, mRNA; Homo sapiens nuclear factor I B (NFIB), transcript variant 33, mRNA; Homo sapiens WT1 transcription factor (WT1), transcript variant 18, mRNA; and Homo sapiens WT1 transcription factor (WT1), transcript variant 19, mRNA. The results are sorted by Default order and are on page 1 of 1467541.

Figure 1.2: Genbank has 18,863,000 sequences under *Homo sapiens*

European Molecular Biology Laboratory (EMBL)

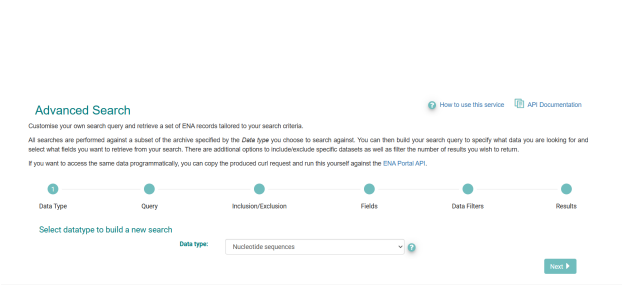


Figure 1.3.1: Querying in EMBL

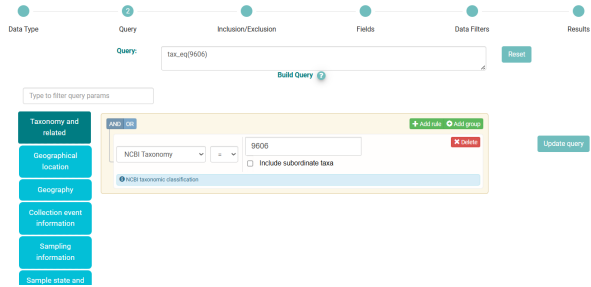


Figure 1.3.2: Querying in EMBL

Download ENA records: FASTA TEXT ?

Download report: JSON TSV ?

Accession	Tax Id
A06851	9606
A08002	9606
A08850	9606
A08905	9606
A11954	9606
A25971	9606
A26249	9606
A26437	9606
A28104	9606
A30438	9606

Items per page: 10 1 - 10 of 10000 (28757917 total) < > >>

Figure 1.3.3: EMBL has 28,757,917 sequences under *Homo sapiens*

Question 2

The nucleotide sequence for AY330867 is obtaining using DDBJ as shown below:

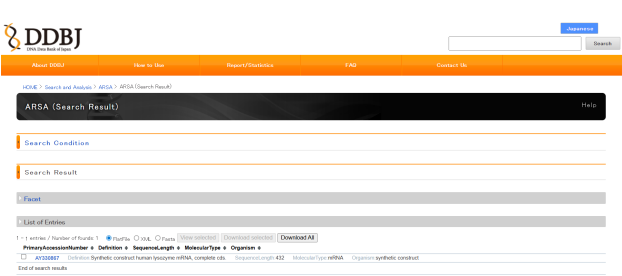


Figure 2.1: Querying in DDBJ

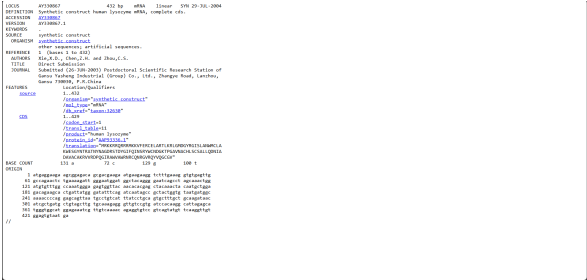


Figure 2.2: The sequence of AY330867

The code to determine the GC (%) content is given below:

```
# Question 2
def find_GC_content(seq):

    GC = 0
    for nt in seq:
        if nt in ("G","C"):
            GC+=1

    GCp = (GC/len(seq))*100
    return(GCp, GC)

seq = "ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAAGGTGTGAGTTGGCCAGAACTCTGAA
AAGATTGGGAATGGATGGCTACAGGGGAATCAGCCTAGCAAACCTGGATGTGTTTGGCCAAATGGGAGAGTGGTACAACACA
CGAGCTACAACTACAATGCTGGAGACAGAAGCACTGATTATGGGATATTTTCAGATCAATAGCCGCTACTGGTGTAAATGAT
GGCAAACCCCAGGAGCAGTTAATGCCTGTCTATTTATCCTGCAGTGTCTTGTCTGCAAGATAACATCGCTGATGCTGTAGCT
TGTGCAAAGAGGGTTGTCCGTGATCCACAAGGCATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAACAGAGGTGTC
CGTCAGTATGTTCAAGGTTGTGGAGTGTAAATGA"
output = find_GC_content(seq)
print("The GC content is:", output[0])
print("The number of GC nucleotides is:", output[1])
print("The total number of nucleotides is:", len(seq))
```

The output generated is given below

```
The GC content is: 46.52777777777778
The number of GC nucleotides is: 201
The total number of nucleotides is: 432
```

Question 3

Although DDBJ, GenBank, and EMBL share most core fields and store equivalent biological information, they differ slightly in their data representation. Among the three, EMBL provides the most comprehensive annotation. Consequently, EMBL entries often present a more detailed and structured view of sequence data.

```
ID AB1598; SV 1; linear, unassigned DNA; PAT; HUM; 423 BP.
XX
AC AB1598;
XX
DT 03-JUL-2002 (Rel. 72, Created)
DT 03-JUL-2002 (Rel. 72, Last updated, Version 1)
XX
DE Human haemoglobin A alpha chain.
XX
KW .
XX
OS Homo sapiens (human)
OC Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia;
OC Eutheria; Euarchontoglires; Primates; Haplorhini; Catarrhini; Hominoidea;
OC Homo.
XX
RN [1]
RP 1-423
RA .
RT "BLOOD SUBSTITUTES";
RL Patent number: W08809179-A1/B, 01-DEC-1988.
XX
DR MDS; tbi1ed4f64cb18fab6d545b7bfe276db.
XX
FH Key Location/Qualifiers
FH
FT source 1..423
FT /organism="Homo sapiens"
FT /mol_type="unassigned DNA"
FT /db_xref="taxon:9606"
FT CDS 1..423
FT /codon_start=1
FT /product="Haemoglobin A alpha chain"
FT /protein_id="CA00102.1"
FT /translation="VLSPADKTNVCAAGVGAGHAGEYGAELERMLSPPTTKYVPH
FT FDLSHGCAQWKGKIVADLLTNAVAHVDPPNALSLDHLKRLVDPVNFLLSKH
FT LLVTLAHLRFETFRANASLOKLAISTVLSKYH"
XX
SQ Sequence 423 BP; 74 A; 156 C; 121 G; 72 T; 0 other;
gtgtctctt ctcctgaaa gccacacgtt aagctcgtt gggc-aaggt tggcgcac 60
gctggcgagt atggtgagga gccctgagag agagtttct tctctctcc caccaccaag 120
acactcttc cgcatttga cctggcgcac ggcctcgcgc aggttaaggg ccaagcgaag 180
aaggttgcgt aagctctgac caagcgtcgt ggcacgtgta adgaagtc caagcgtg 240
ctgcctgta ggcacgtgca cgcgcacag cttcgagtg acccgtcaa cttaagctc 300
ctgaagcct gctcgtgat ggcctcgtg gccacctc cgcgcgatt cacccttgc 360
gtgcagcct ccttgagaaa gttctggtt tctgtgagca cgtgctgac ctccaatac 420
cgt
//
```

Figure 3.1: EMBL Example 1

```
ID AB1592; SV 1; linear, unassigned DNA; PAT; HUM; 438 BP.
XX
AC AB1592;
XX
DT 08-MAR-1993 (Rel. 35, Created)
DT 08-MAR-1993 (Rel. 35, Last updated, Version 1)
XX
DE Human haemoglobin A beta chain.
XX
KW .
XX
OS Homo sapiens (human)
OC Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia;
OC Eutheria; Euarchontoglires; Primates; Haplorhini; Catarrhini; Hominoidea;
OC Homo.
XX
RN [1]
RP 1-438
RA .
RT "BLOOD SUBSTITUTES";
RL Patent number: W08809179-A1/B, 01-DEC-1988.
XX
DR MDS; bdl7aac3d8aca9989f875a49d53e708c.
XX
FH Key Location/Qualifiers
FH
FT source 1..438
FT /organism="Homo sapiens"
FT /mol_type="unassigned DNA"
FT /db_xref="taxon:9606"
FT CDS 1..438
FT /codon_start=1
FT /product="haemoglobin A beta chain"
FT /protein_id="CA00102.1"
FT /translation="VLTIESAVTALMGVMDVGGALGRLLVVPMTQFFESF
FT GDLSTDAVMKPKVAGKXVLGASGLAHLNMLKGTATLSLHCKDLHPENFR
FT LLGWLVCVLRHFGGEEFTPPVQAYQKVAGVANAIAKHY"
XX
SQ Sequence 438 BP; 85 A; 114 C; 135 G; 104 T; 0 other;
gtgcactta cttctgaga aagcttgcg gttacttcc ttatgaaa aaatgaagtg 60
gatgaattg gttgtgagg cttggcagg cttgtgaga tctacctg gccccagg 120
tctttgagc cttttggga ttgctccc cttgagtg ttatggca ccttcagg 180
aaggtcttg caaaaagt gctgggct ttttgtgta gcttgctca cttgacac 240
ctcaaggca ctttgcac acgtgtag cttacatga acagctga cttgacct 300
gagactta gctctctg caactctg gttctgag tggccaca cttggaaa 360
gaattcacc caccagtgca ggtctctat cagaaatgg tggctggt ggttaagcc 420
ctggccaca agtatcac
//
```

Figure 3.2: EMBL Example 2

```
LOCUS       AB1590               423 bp    DNA    linear    PAT 03-JUL-2002
DEFINITION  Human haemoglobin A alpha chain.
ACCESSION   AB1590
VERSION     AB1590.1
KEYWORDS    -
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorhini;
            Catarrhini; Hominidae; Homo.
            1 (bases 1 to 423)
REFERENCE   1
  AUTHORS   BLOOD SUBSTITUTES
  TITLE     Patent number M08B9179-A1/8, 01-DEC-1988.
  JOURNAL   Location/Qualifiers
  FEATURES  source
            1..423
            /organism="Homo sapiens"
            /mol_type="unassigned DNA"
            /db_xref="taxon:9606"
            c1..423
            /codon_start=1
            /product="Haemoglobin A alpha chain"
            /protein_id="CA000102.1"
            /translation="VLPADKTNWAAKGKVGAGHGEYGAELRMFLSPFTKTYTF
            HQLSHGSAQWKGKGVADALTNAAHVRDOPHALSALSDLHAKLRVDPVNFKLLS
            HCLLYLAAHFAETTPAHNSLKLADSVTSKYR"
BASE COUNT  74 a      156 c      121 g      72 t
ORIGIN
1 ggtgtgtctt ctgcgcacaa gccacacgtc aagacgcctt ggggcacagt tggcgcac
61 gctgcagagt atggtgcagg ggccttgagb agatgttct tctctctcc caccaccaag
121 actctcttc cgcctcttgc ctgcgcacaa gctctctcc aggttcagag cccgcgcagg
181 aagatgcagg acgcgtgac caacgcctgg ggcacgtgag agcacatgcc caagcgcgtg
241 tgcgcctgag agacgtcga cgcgcacaa ctttggatgg accgatcaga cttcagcttc
301 cttaagccat gctgttgtt gacctctgc gcccaacttc cgcgcctggt caccctctgg
361 gtagcagctt ccttgacaaa gttcttgct tcttgagca cgtgtgacg ctccaatcc
423 cgt
//
```

Figure 3.3: DDBJ Example 1

```
LOCUS       AB1592               438 bp    DNA    linear    PAT 08-MAR-1993
DEFINITION  Human haemoglobin A beta chain.
ACCESSION   AB1592
VERSION     AB1592.1
KEYWORDS    -
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorhini;
            Catarrhini; Hominidae; Homo.
            1 (bases 1 to 438)
REFERENCE   1
  AUTHORS   BLOOD SUBSTITUTES
  TITLE     Patent number M08B9179-A1/10, 01-DEC-1988.
  JOURNAL   Location/Qualifiers
  FEATURES  source
            1..438
            /organism="Homo sapiens"
            /mol_type="unassigned DNA"
            /db_xref="taxon:9606"
            c1..438
            /codon_start=1
            /product="Haemoglobin A beta chain"
            /protein_id="CA000102.1"
            /translation="VHLTPREKSAVTALGKGVNDEVGAGLGLVVPMTHQFFES
            FQSLSTPWAWPVPVKAHKKVGLGASGLAHDLNKGATATLSELKDLNDFPFI
            FLGLNLVCLVAHFHGFETPPVQAQVQVQVWALAHKHYR"
BASE COUNT  85 a      154 c      125 g      104 t
ORIGIN
1 gttgccttgc ctcttgagaa gaagtcacac ctgtctgac tatggcaca agtgcagtg
61 gttgcagtg gttgttggc cttgcgcagg cttgtctgc tctaccttg gccccagagg
121 ttttttgagt cttttggaga ttgtctcact cttgtatgc ttatggaaa ccttcagagg
181 aagctctgc ggcacagagt gttgcagtc cttgtatgc ttatggaaa ccttcagagg
241 ctacagagca cttttgcac actgcagtag ctgcacttg accgtatgca ctgagctct
301 gaaagcttca agttctttag cagatctgac atctgtatg tggcctatca ctttgagaa
361 gaattccccc caccatgca agctctcat cagaaatgag ttgcttatgt agttatgac
421 ctggccaca agtatcac
//
```

Figure 3.4: DDBJ Example 2

GenBank → Send to →

Human haemoglobin A alpha chain

GenBank: AB1590.1

FASTA Graphics

Go to: [GenBank](#)

LOCUS AB1590 423 bp DNA linear PAT 03-JUL-2002

DEFINITION Human haemoglobin A alpha chain.

ACCESSION AB1590

VERSION AB1590.1

KEYWORDS -

SOURCE Homo sapiens (human)

ORGANISM Homo sapiens

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;

Mammalia; Eutheria; Euarchontoglires; Primates; Haplorhini;

Catarrhini; Hominidae; Homo.

1 (bases 1 to 423)

REFERENCE 1

AUTHORS BLOOD SUBSTITUTES

TITLE Patent: NO 88B9179-A1 8 01-DEC-1988;

JOURNAL Location/Qualifiers

FEATURES source

1..423

/organism="Homo sapiens"

/mol_type="unassigned DNA"

/db_xref="taxon:9606"

c1..423

/codon_start=1

/product="Haemoglobin A alpha chain"

/protein_id="CA000102.1"

/translation="VLPADKTNWAAKGKVGAGHGEYGAELRMFLSPFTKTYTF

HQLSHGSAQWKGKGVADALTNAAHVRDOPHALSALSDLHAKLRVDPVNFKLLS

HCLLYLAAHFAETTPAHNSLKLADSVTSKYR"

ORIGIN

1 ggtgtgtctt ctgcgcacaa gccacacgtc aagacgcctt ggggcacagt tggcgcac

61 gctgcagagt atggtgcagg ggccttgagb agatgttct tctctctcc caccaccaag

121 actctcttc cgcctcttgc ctgcgcacaa gctctctcc aggttcagag cccgcgcagg

181 aagatgcagg acgcgtgac caacgcctgg ggcacgtgag agcacatgcc caagcgcgtg

241 tgcgcctgag agacgtcga cgcgcacaa ctttggatgg accgatcaga cttcagcttc

301 cttaagccat gctgttgtt gacctctgc gcccaacttc cgcgcctggt caccctctgg

361 gtagcagctt ccttgacaaa gttcttgct tcttgagca cgtgtgacg ctccaatcc

423 cgt

//

Figure 3.5: Genbank Example 1

GenBank → Send to →

Human haemoglobin A beta chain

GenBank: AB1592.1

FASTA Graphics

Go to: [GenBank](#)

LOCUS AB1592 438 bp DNA linear PAT 08-MAR-1993

DEFINITION Human haemoglobin A beta chain.

ACCESSION AB1592

VERSION AB1592.1

KEYWORDS -

SOURCE Homo sapiens (human)

ORGANISM Homo sapiens

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;

Mammalia; Eutheria; Euarchontoglires; Primates; Haplorhini;

Catarrhini; Hominidae; Homo.

1 (bases 1 to 438)

REFERENCE 1

AUTHORS BLOOD SUBSTITUTES

TITLE Patent: NO 88B9179-A1 10 01-DEC-1988;

JOURNAL Location/Qualifiers

FEATURES source

1..438

/organism="Homo sapiens"

/mol_type="unassigned DNA"

/db_xref="taxon:9606"

c1..438

/codon_start=1

/product="Haemoglobin A beta chain"

/protein_id="CA000102.1"

/translation="VHLTPREKSAVTALGKGVNDEVGAGLGLVVPMTHQFFES

FQSLSTPWAWPVPVKAHKKVGLGASGLAHDLNKGATATLSELKDLNDFPFI

FLGLNLVCLVAHFHGFETPPVQAQVQVQVWALAHKHYR"

ORIGIN

1 gttgccttgc ctcttgagaa gaagtcacac ctgtctgac tatggcaca agtgcagtg

61 gttgcagtg gttgttggc cttgcgcagg cttgtctgc tctaccttg gccccagagg

121 ttttttgagt cttttggaga ttgtctcact cttgtatgc ttatggaaa ccttcagagg

181 aagctctgc ggcacagagt gttgcagtc cttgtatgc ttatggaaa ccttcagagg

241 ctacagagca cttttgcac actgcagtag ctgcacttg accgtatgca ctgagctct

301 gaaagcttca agttctttag cagatctgac atctgtatg tggcctatca ctttgagaa

361 gaattccccc caccatgca agctctcat cagaaatgag ttgcttatgt agttatgac

421 ctggccaca agtatcac

//

Figure 3.6: Genbank Example 2

Field	DDBJ	Genbank	EMBL
Locus	Present	Present	Absent
ID	Absent	Absent	Present
Definition	Present	Present	Present
Accession	Present	Present	Present
Version	Present	Present	Present (in DT)
Keywords	Present	Present	Present
Source Organism	Present	Present	Present (in OS)
Reference	Present	Present	Present (in RN and RP)
Features	Present	Present	Present
Translation	Present	Present	Present
Base Count	Present	Absent	Present
Sequence	Present	Present	Present

Question 4

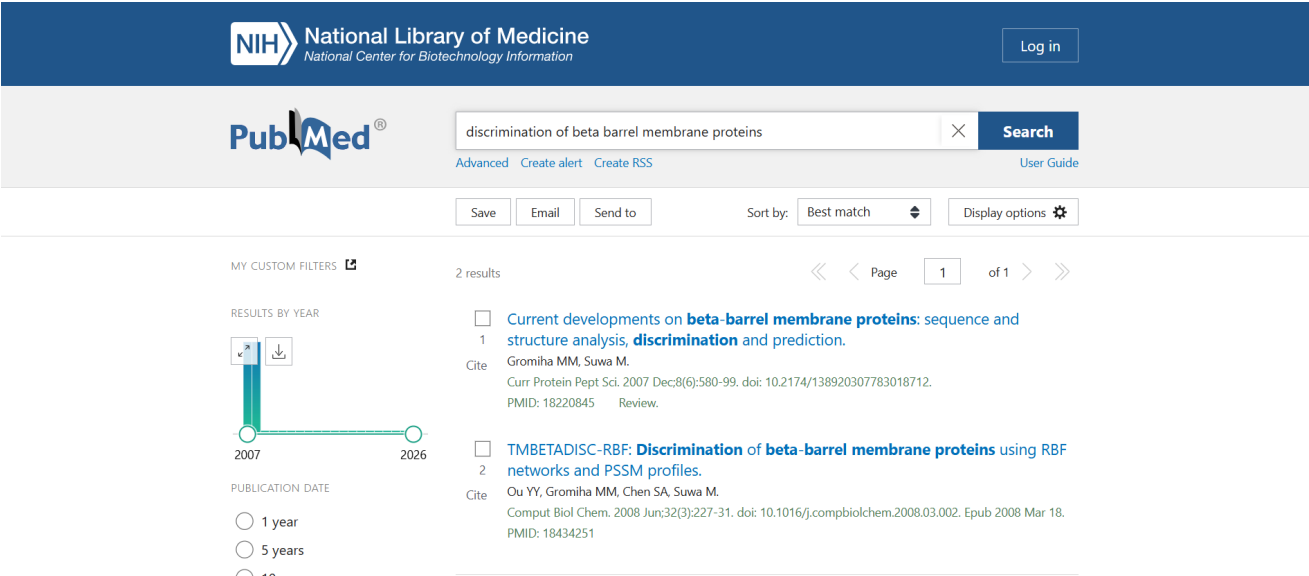


Figure 4.1: Papers on “discrimination of beta barrel membrane proteins”

Question 5

Using Google Scholar to find the number of papers written by Daisuke Kihara.

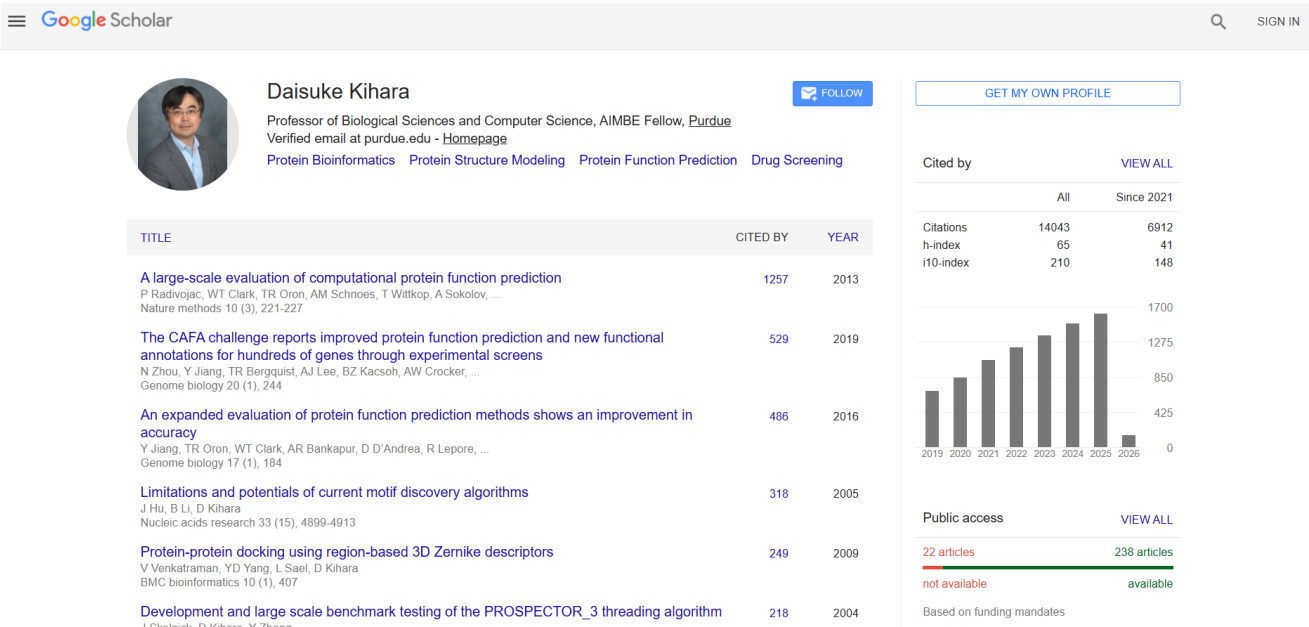


Figure 5.1: 260 Papers have been published by Daisuke Kihara

Using PubMed to find the number of papers written by Daisuke Kihara.

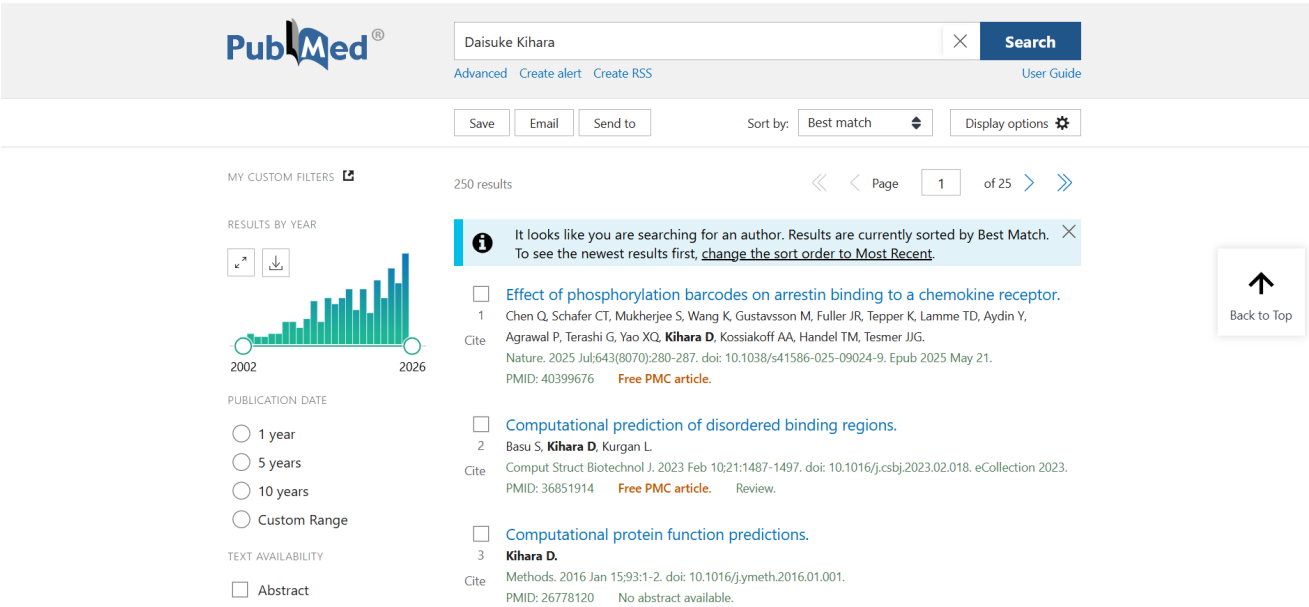


Figure 5.2: 250 Papers have been published by Daisuke Kihara according to PubMed

Question 6

The set of *Related Articles* constitutes of *similar articles* and *articles that cite the main article*.



Figure 6.1: PMID of the original paper

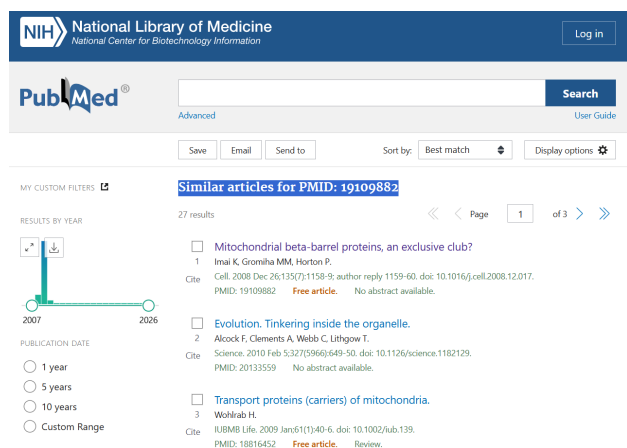


Figure 6.2: 27 Similar Articles

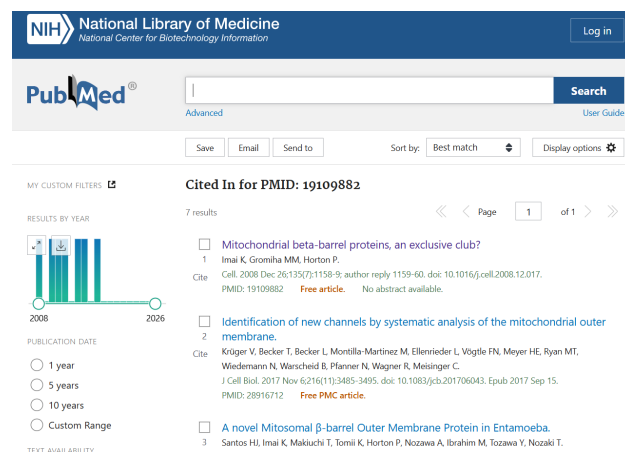


Figure 6.3: 7 Articles that cite the Main Article

The **34 related articles** are listed shown below according to which class they belong to.

Question 7

NOTE: For the purpose of this query, the ISSN 0028-0836 was used, which uniquely identifies the print version of the journal Nature. The Online/Electronic ISSN (eISSN: 1476-4687) was not considered for this query.



Figure 7.1: 4177 Articles have been published in the journal "Nature" in the year 2024 according to PubMed

Advanced query ☐

ISSN (0028-0836) AND PUBYEAR = 2024

Show less

Save search

Set search alert

Edit in advanced search

Documents Preprints Beta Secondary documents

3,427 documents found

Analyze results

Refine search

Search within results

Filters

Year

☒ Range ☐ Individual☐ All ☐ Export ☐ Download ☐ Citation overview ☐ MoreShow all abstracts Sort by Date (newest) ☐ ☐

	Document title	Authors	Source	Year	Citations
--	----------------	---------	--------	------	-----------

☐ 1	Article Atmospheric rivers cause warm winters and extreme heat events	Scholz, S.R., Lora, J.M.	Nature, 636(8043), pp. 640–646	2024	21
----------------------------	---	--------------------------	--------------------------------	------	----

Show abstract ☐ Full Text ☐ View at Publisher ☐ Related documents

Figure 7.2: 3427 Articles have been published in the journal "Nature" in the year 2024 according to Scopus

Question 8

Burkhard Rost has a h-index of 116 and 62,265 citations.

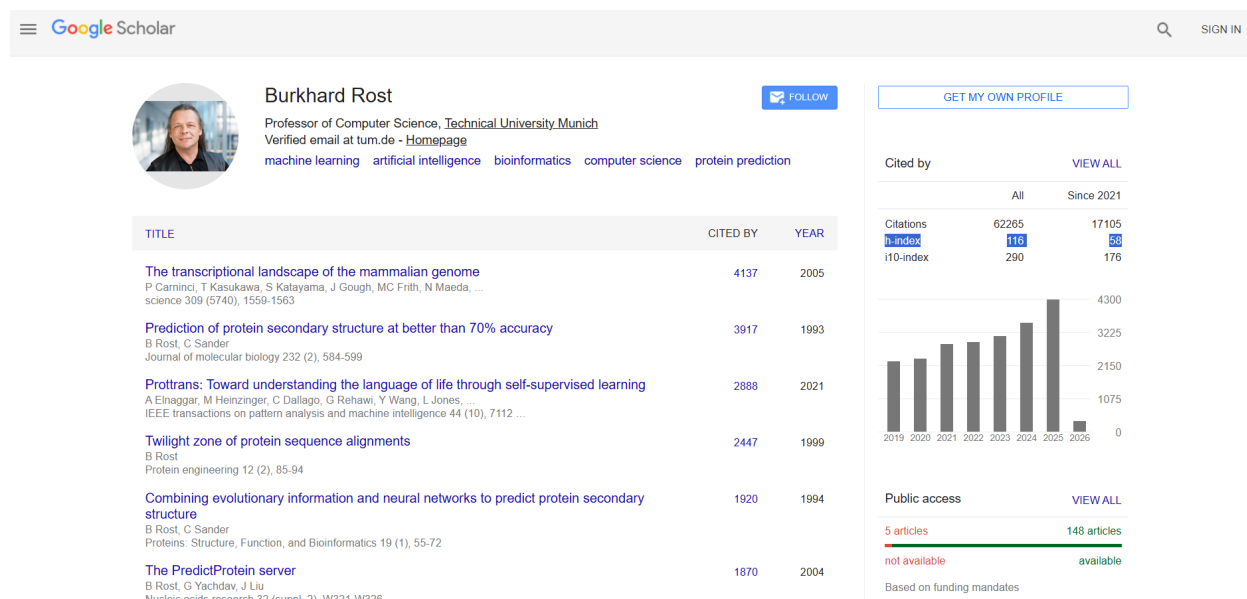


Figure 8.1: Google Scholar Page of Burkhard Rost

Question 9

The enzyme is classified as given below:

1. EC 1 : Oxidoreductase (catalyze redox reactions)
2. EC 1.7 : Acting on other nitrogenous compounds as donors
3. EC 1.7.2 : With a cytochrome as acceptor
4. EC 1.7.2.3 : Trimethylamine-N-oxide reductase

The Catalytic Site Data obtained from Catalytic Site Atlas for **EC 6.3.5.4**.

Catalytic Residues Roles			
UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

*PDB label guide - RESx(y)B(C) - RES: Residue Name; x: Residue ID in PDB file; y: Residue ID in PDB sequence if different from PDB file; B: PDB Chain; C: Biological Assembly Chain if different from PDB. If label is "Not Found" it means this residue is not found in the reference PDB.

Figure 10.3: Identifying Enzyme Catalytic Sites using Catalytic Site Atlas

Question 11

Organism	Scientific Name	Taxonomy ID	No. of Chromosomes
Human	<i>Homo sapiens</i>	9606	46 (22 autosomal pairs, X, Y)
Cat	<i>Felis catus</i>	9685	38 (18 autosomal pairs, X, Y)
Dog	<i>Canis lupus familiaris</i>	9615	78 (38 autosomal pairs, X, Y)
Domestic guinea pig	<i>Cavia porcellus</i>	10141	64 (31 autosomal pairs, X, Y)
Thale cress	<i>Arabidopsis thaliana</i>	3702	10 (5 autosomal pairs, doesn't have allosomes)

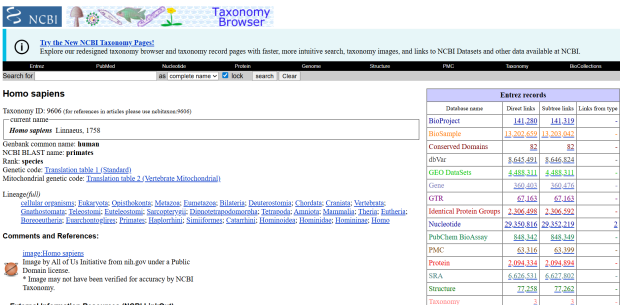


Figure 11.1: Human; Scientific Name, Taxonomy ID

Figure 11.2: Human; Number of Chromosomes

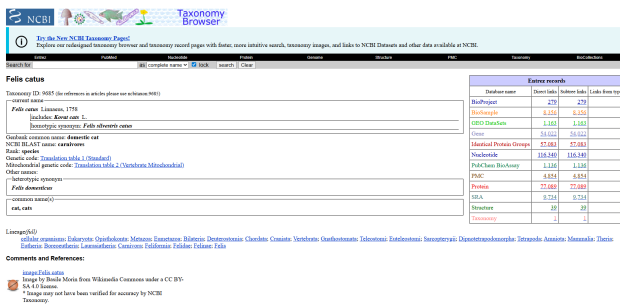


Figure 11.3: Cat; Scientific Name, Taxonomy ID

Chromosomes



Figure 11.4: Cat; Number of Chromosomes

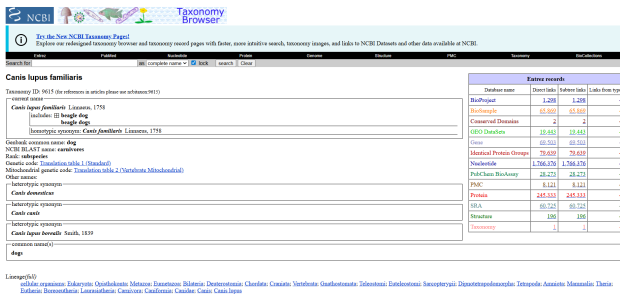


Figure 11.5: Dog; Scientific Name, Taxonomy ID

Chromosomes

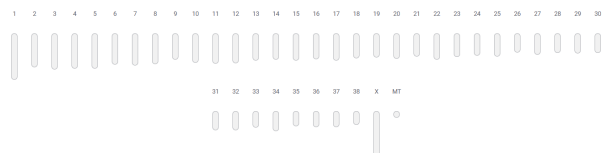


Figure 11.6: Dog; Number of Chromosomes

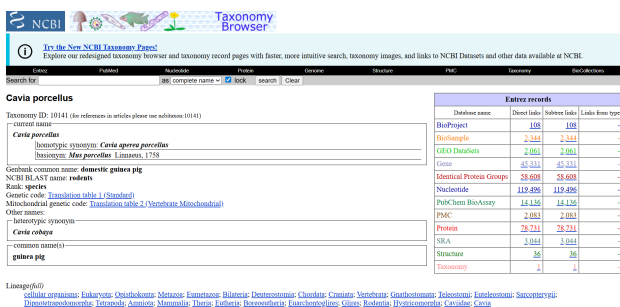


Figure 11.7: Domestic Guinea Pig; Scientific Name, Taxonomy ID

Chromosomes



Figure 11.8: Domestic Guinea Pig; Number of Chromosomes

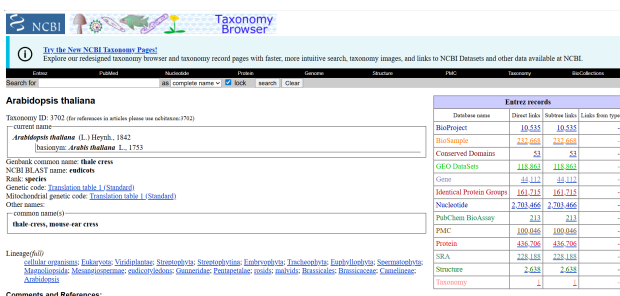


Figure 11.9: Thale Cress; Scientific Name, Taxonomy ID

Chromosomes



Figure 11.10: Thale Cress; Number of Chromosomes

NOTE: The guinea pig genome was sequenced in 2008 as part of the Mammalian Genome Project, but the guinea pig sequence scaffolds have not been assigned to chromosomes.

Accordingly, karyotype analysis of *Cavia porcellus* revealed a diploid chromosome number ($2n = 64$), comprising 62 autosomes and 2 sex chromosomes.

Question 12

Efetch is an NCBI E-Utility that is used to download full data records (like DNA sequences or PubMed abstracts) from Entrez databases in specific formats such as FASTA or XML.

https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=<database>&id=<uid_list>&rettype=<retrieval_type>&retmode=<retrieval_mode>

Example: Download nucleotide GIs 34577062 and 24475906 in FASTA format

<https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nucleotide&id=34577062,24475906&rettype=fasta&retmode=text>

Question 13

Three database examples for each case have been listed below:

S.No	Protein Properties	Small Molecules	Cancer Genes
1	AAindex	BitterDB	ArrayMap
2	BindingDB	ChemBank	Cancer3D
3	Cybase	ChemDB	CancerGenes

Protein properties

AAindex
BindingDB
Cybase
dbPTM

Small molecules

BitterDB
ChEBI - Chemical Entities of Biological Interest
ChemBank
ChemDB
Crystallography Open Database
CSD - Cambridge Structural Database

Figure 13.1: Protein Property Databases

Figure 13.2: Small Molecule Databases

Cancer gene databases

ArrayMap
BCCTBbp
BreCAN-DB
Cancer RNA-Seq Nexus
Cancer3D
CancerGenes
CancerPPD
Candidate Cancer Gene Database
CanGEM

Figure 13.3: Cancer Gene Databases

Link to the google drive containing the python code used in this assignment: <https://drive.google.com/drive/folders/1qaGKvJXPJvfi0nQg2Au0dhZao6a7SHfq?usp=sharing>